

SER216 — Software Engineering

Final Practical Project

Digital Parking Permit System

Task 1: Requirements

1.1 Functional Requirements

Functional requirements describe the specific behaviors the system must perform.

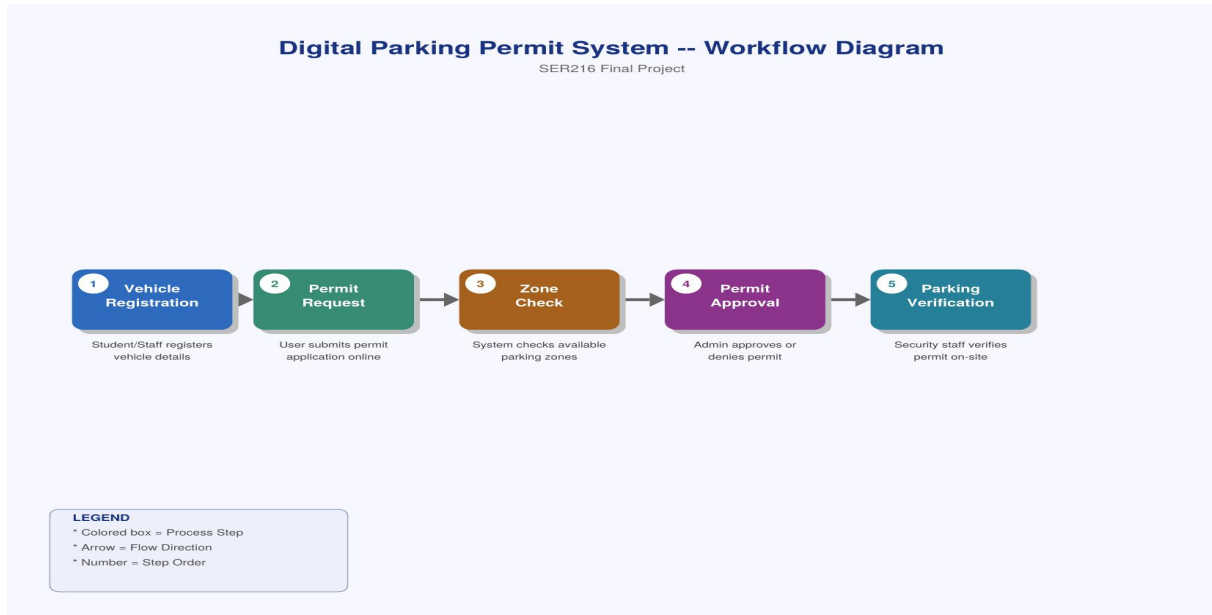
FR-01	Vehicle Registration	The system shall allow registered students and staff to add, edit, and remove vehicle information including plate number, make, model, and year.
FR-02	Parking Permit Request	The system shall allow eligible users to submit a parking permit application for a selected zone. The application must be reviewed and approved or denied by an administrator.
FR-03	Zone Availability Check	The system shall display real-time availability of parking zones and allow users to filter zones by type (student, staff, accessible) before submitting a permit request.
FR-04	Parking Issue Reporting	The system shall provide a reporting interface for students and staff to submit parking-related issues. Security staff must be notified automatically upon submission.

1.2 Non-Functional Requirements

NFR-01	Performance	The system shall process permit requests and display zone availability within 3 seconds under normal load conditions (up to 500 concurrent users).
NFR-02	Security	All user data must be encrypted in transit using TLS 1.2 or higher. Access to administrative functions must require role-based authentication.

Task 2: Workflow Diagram

The workflow below illustrates the end-to-end process of the Digital Parking Permit System, created using diagrams.net.



Workflow Steps:

- 1. Vehicle Registration:** Student/Staff logs in and registers their vehicle details.
- 2. Permit Request:** The user selects a parking zone and submits a permit application online.
- 3. Zone Check:** The system verifies zone availability and user eligibility automatically.
- 4. Permit Approval:** An administrator reviews and approves or denies the application.
- 5. Parking Verification:** Security staff verify the digital permit when the user parks.

Task 3: Test Case Table

6 test cases covering System Testing (2), Acceptance Testing (2), and Regression Testing (2). Full table available as test_cases.xlsx.

Test ID	Feature	Input / Data	Expected Result	Test Type
TC-001	Vehicle Registration	Valid student ID, vehicle plate, model	Vehicle registered; confirmation shown	System Test
TC-002	Permit Approval Workflow	Admin receives complete permit request	Request approved; status Active; email sent	System Test
TC-003	Parking Permit Request	Registered user requests permit	Permit confirmed and zone assigned	Acceptance Test
TC-004	Parking Issue Reporting	Student reports broken gate at Zone B	Issue logged; ticket generated; security notified	Acceptance Test
TC-005	Vehicle Registration After Update	Valid registration submitted post-update	Registration works correctly — no regression	Regression Test
TC-006	Permit Expiry Verification	Security verifies previously-expired permit	Permit correctly shows Active after fix	Regression Test

Task 4: Defect Analysis

Defect: "A valid parking permit is shown as expired during security verification."

Defect Category	Data Display / Functional Defect	The system displays incorrect permit status. The permit is valid but the UI shows it as expired.
Severity	High	Prevents security staff from correctly verifying permits; legitimate users denied access.
Priority	High	Must be fixed immediately as it directly affects daily operations and user trust.
Possible Root Cause	Date/Time Logic Error	The expiry date comparison likely uses an incorrect format, time zone, or off-by-one error.
Regression Test	TC-006	After the fix, verify a known-valid permit shows as Active. Also test with a genuine expired permit.

Task 5: GitHub Evidence

5.1 GitHub Repository Link

URL: https://github.com/YourUsername/SER216_Final_Project_StudentName

5.2 Completed GitHub Tasks

Task	Details	Done
Create Repository	SER216_Final_Project_StudentName	Yes
README.md	Describes project, tools, files, workflow	Yes
Upload Files	Report, diagram, test table, screenshots	Yes
Create Branch	feature/test-plan	Yes
3+ Commits	Meaningful commit history	Yes
Pull Request	From feature/test-plan to main	Yes
Merge PR	Merged after review	Yes
GitHub Link in PDF	See Section 5.1 above	Yes

5.3 Sample Commit Messages

```
git commit -m "Add project report structure"
git commit -m "Add workflow diagram"
git commit -m "Add test case table"
git commit -m "Add defect analysis"
git commit -m "Update README and final PDF"
```

5.4 Required Screenshot Evidence

Include these screenshots in the screenshots/ folder and this PDF:

- * repo_homepage.png – Repository main page
- * commits.png – Commit history (3+ commits)
- * branch.png – Feature branch view
- * pull_request.png – Pull request (open and merged)